JEE Mathematics: Quick Review with Grading and Practice Map - Method of Differentiation (Batch XIII)

Grade shows how often the item is tested in the JM exercise of this batch, the JA exercise and the JEE Main Question Bank taken together (33 questions; a question that appears in two of these is counted once): High = 3 or more questions, Medium = 1 to 2 questions, Low = none. Low only means the item is not tested in these questions; it can still be important for understanding. Practice map lists the questions that test the item in the sheet exercises (J-I, J-II, J-M, J-A, C-I, C-II), the DPPs and the Question Bank (QB). 'Main concept' = the question is mainly about this item; 'Also used in' = the question needs it as a secondary step.

Differentiation by first principle

Question 01

How to differentiate a function by first principle?

Model Answer

$$\frac{dy}{dx} = f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

It is the instantaneous rate of change of f (also called the delta / ab-initio method). A limit of this shape is a derivative in disguise.

Ex: $f = x^2$: $\lim_{h\to 0} \frac{(x+h)^2 - x^2}{h} = \lim_{h\to 0}(2x + h) = 2x$.

Grade - Medium

Practice map, main concept: Sheet: J-I 45; J-M 20; C-I 15, 19; C-II 7

Also used in: Sheet: J-I 50; J-II 22

Question 02

What does a dot at the top of a function represent?

Model Answer

A dot means the derivative with respect to time: $\dot{S} = \frac{dS}{dt}$, $\dot{\theta} = \frac{d\theta}{dt}$.

For $y = f(x)$ the symbols $\frac{dy}{dx}$, Dy, $f'(x)$, y_1, y' all mean the same.

Grade - Low

Practice map, main concept: none

Also used in: none

Derivative of standard function

Question 03

Write the derivatives of all 19 standard functions.

Model Answer

$$(c)' = 0,\ (ax)' = a,\ (x^n)' = nx^{n-1},\ (e^x)' = e^x,\ (a^x)' = a^x \ln a$$

$$(\ln x)' = \frac{1}{x}, \quad (\log_a x)' = \frac{1}{x \ln a}$$

$$(\sin x)' = \cos x,\ (\cos x)' = -\sin x,\ (\tan x)' = \sec^2 x,\ (\cot x)' = -\operatorname{cosec}^2 x$$

$$(\sec x)' = \sec x \tan x, \quad (\operatorname{cosec} x)' = -\operatorname{cosec} x \cot x$$

$$(\sin^{-1} x)' = \frac{1}{\sqrt{1-x^2}} = -(\cos^{-1} x)', \quad -1 < x < 1$$

$$(\tan^{-1} x)' = \frac{1}{1+x^2} = -(\cot^{-1} x)'$$

$$(\sec^{-1} x)' = \frac{1}{|x|\sqrt{x^2-1}} = -(\operatorname{cosec}^{-1} x)', \quad |x| > 1$$

Ex: $\frac{d}{dx}(2^x e^x) = \frac{d}{dx}(2e)^x = (2e)^x \ln(2e)$.

Grade - High

Practice map, main concept: Sheet: J-I 2, 23; J-II 45; J-M 3, 9; C-I 11, 16, 18; C-II 23

Also used in: Sheet: J-I 1, 8, 11, 16, 28-29; J-II 16, 44; J-M 4, 16, 20, 23; J-A 1; C-I 20, 22; C-II 2, 21 | Question Bank: 138

Important Theorems

Question 04

Write the product rule of differentiation.

Model Answer

$$(fg)' = f'g + fg', \qquad (fgh)' = f'gh + fg'h + fgh'$$

Also $(fgh)' = \frac{1}{2}\big[(fg)'h + (gh)'f + (hf)'g\big]$; $(f \pm g)' = f' \pm g'$, $(kf)' = kf'$.

Ex: $f(0) = 1, g(0) = 2, h(0) = 3, (fg)'(0) = 6, (gh)'(0) = 4, (hf)'(0) = 5 \Rightarrow (fgh)'(0) = \frac{18+4+10}{2} = 16$.

Grade - High

Practice map, main concept: Sheet: J-I 12; J-II 1; J-M 10; J-A 1, 5; C-I 20-21; C-II 20-22 | Question Bank: 135, 138

Also used in: Sheet: J-I 50; J-II 3, 42; J-M 2, 21-22; C-I 15; C-II 4

Question 05

Write the quotient rule of differentiation.

Model Answer

$$\frac{d}{dx}\left(\frac{f}{g}\right) = \frac{g\,f' - f\,g'}{g^2}, \quad g \neq 0$$

Ex: $\frac{d}{dx}\frac{x}{\sin x} = \frac{\sin x - x\cos x}{\sin^2 x}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 11; J-II 14-15; C-I 2 | Question Bank: 143

Also used in: Sheet: J-I 5, 29, 38; J-II 8; J-M 8

Question 06

What to keep in mind before applying the quotient rule?

Model Answer

First simplify: factorise and cancel, use trig / ITF identities, divide out. Often the quotient becomes a simple function.

Ex: $\frac{x^4+x^2+1}{x^2+x+1} = x^2 - x + 1 \Rightarrow y' = 2x - 1$; $\frac{\tan^2 x \sin^2 x}{\tan^2 x - \sin^2 x} = 1 \Rightarrow y' = 0$.

Grade - High

Practice map, main concept: Sheet: J-I 6, 26; J-II 7, 16, 44; J-M 16; C-I 4, 22; C-II 13 | Question Bank: 134, 146

Also used in: Sheet: J-I 7, 42; J-II 9, 18-19, 23; J-M 2-3, 8, 11, 19; C-I 21; C-II 23 | Question Bank: 143

Question 07
Write the chain rule.

Model Answer

$$\frac{d}{dx} f(g(x)) = f'(g(x))\, g'(x), \qquad \frac{dy}{dx} = \frac{dy}{dz} \cdot \frac{dz}{dx}$$

Ex: $\frac{d}{dx} e^{x^2} = 2x\, e^{x^2}$.

Grade - High

Practice map, main concept: Sheet: J-I 1, 3, 8, 29; J-II 3; J-M 4; J-A 3-4

Also used in: Sheet: J-I 10, 33-34, 41, 43, 48; J-II 7, 14-15, 31, 40; J-M 3, 11, 16, 19, 21, 24; J-A 5; C-I 1, 4, 9, 12; C-II 5, 8, 16-17 | Question Bank: 137, 139, 146

Question 08
What is the relation between $\frac{dy}{dx}$ and $\frac{dx}{dy}$?

Model Answer

$$\frac{dy}{dx} = \frac{1}{dx/dy}, \quad \frac{dx}{dy} \neq 0$$

From the chain rule: $\frac{dy}{dx} \cdot \frac{dx}{dy} = \frac{dy}{dy} = 1$.

Ex: $x = y + y^3 \Rightarrow \frac{dx}{dy} = 1 + 3y^2 \Rightarrow \frac{dy}{dx} = \frac{1}{1+3y^2}$.

Grade - Low

Practice map, main concept: Sheet: J-I 5

Also used in: Sheet: J-I 46; C-II 3

Question 09
How to differentiate a composite function with the help of the chain rule?

Model Answer

Peel from outside to inside and multiply the derivatives of every layer: $\frac{d}{dx} f(g(h(x))) = f'(g(h))\, g'(h)\, h'(x)$.

Ex: $f'(x) = \sin(x^2)$, $y = f(x^2+1)$: $\frac{dy}{dx} = 2x \sin\left((x^2+1)^2\right)$; at $x = 1$: $2 \sin 4$.

Grade - Medium

Practice map, main concept: Sheet: J-I 27, 32; J-M 22; C-I 5, 7; C-II 4, 8, 25

Also used in: Sheet: J-I 35; J-A 2; C-I 17 | Question Bank: 135

Logarithmic Differentiation

Question 10
When to apply logarithmic differentiation?

Model Answer

(i) y is a product or quotient of many factors; (ii) $y = [f(x)]^{g(x)}$ (variable base and variable power).

Ex: $x^{\sin x}$, x^x, $(\sin x)^{\ln x}$, $\frac{\sqrt{x}(1-2x)^{2/3}}{(2-3x)^{3/4}(3-4x)^{4/5}}$.

Grade - Medium

Practice map, main concept: none

Also used in: Sheet: J-I 14; J-II 2, 5, 10, 20; J-M 14; C-I 23-24 | Question Bank: 142

Question 11
How to apply logarithmic differentiation?

Model Answer
Take ln of both sides, expand with log rules, differentiate: $\frac{1}{y}\frac{dy}{dx} = \ldots$, then multiply by y.

Ex: $f = (x+1)(x+2)\cdots(x+n)$: $\frac{f'}{f} = \sum \frac{1}{x+r} \Rightarrow f'(0) = n!\left(1 + \frac{1}{2} + \cdots + \frac{1}{n}\right)$.

Grade - **High**
Practice map, main concept: Sheet: J-I 9, 14-15; J-II 2, 4-5, 10, 20; J-M 14; C-I 23-24 | Question Bank: 142
Also used in: Sheet: J-I 18-19, 21, 24-25, 40; J-II 41; J-M 5, 7 | Question Bank: 144

Question 12
What is another method to differentiate $(f(x))^{g(x)}$?

Model Answer

$$\frac{d}{dx} f^{\,g} = g\, f^{\,g-1} f' + f^{\,g} \ln f \cdot g' = f^{\,g}\left[g' \ln f + \frac{g\, f'}{f}\right]$$

Differentiate as a power (exponent constant) plus as an exponential (base constant), and add; or write $f^{\,g} = e^{g \ln f}$.

Ex: $\frac{d}{dx} x^{\sin x} = x^{\sin x}\left[\cos x \ln x + \frac{\sin x}{x}\right]$.

Grade - **Medium**
Practice map, main concept: Sheet: J-M 5
Also used in: Sheet: J-I 4, 13; J-II 36, 38; J-M 7; C-I 23

Parametric Differentiation

Question 13
How to find the derivative for a curve which is given parametrically?

Model Answer

$$x = g(t),\ y = f(t): \quad \frac{dy}{dx} = \frac{dy/dt}{dx/dt}$$

Ex: $x = a\cos^3 t,\ y = a\sin^3 t$: $\frac{dy}{dx} = \frac{3a\sin^2 t \cos t}{-3a\cos^2 t \sin t} = -\tan t$.

Grade - **Low**
Practice map, main concept: Sheet: J-I 22; J-II 6, 8
Also used in: Sheet: J-II 27; C-I 3, 17

Derivative of one function wrt another function

Question 14
How to find the derivative of one function with respect to another function?

Model Answer

$$\frac{d\, f(x)}{d\, g(x)} = \frac{f'(x)}{g'(x)}$$

Treat x as a parameter. With ITFs, first simplify both by a substitution ($x = \tan\theta$, $x = \sin\theta$ ...).

Ex: $\sin x$ w.r.t. $\cos x$: $\frac{\cos x}{-\sin x} = -\cot x$.

Grade - **Low**
Practice map, main concept: Sheet: J-I 17; J-II 9, 17, 23; C-I 13
Also used in: Sheet: J-II 46

Derivative of implicit functions

Question 15

How to find the derivative of a function which is given in implicit form?

Model Answer

Differentiate every term w.r.t. x (treat y as a function of x: $\frac{d}{dx}y^2 = 2y\,y'$), collect the y' terms and solve. If possible, simplify the relation first.

$$f(x,y) = 0 \Rightarrow \frac{dy}{dx} = -\frac{\partial f/\partial x}{\partial f/\partial y}$$

Ex: $y^5 + xy^2 + x^3 = 4x + 3$ at $(2,1)$: $(5+4)y' + 1 + 12 = 4 \Rightarrow y' = -1$.

Grade - Medium

Practice map, main concept: Sheet: J-I 4, 7, 13, 21; J-II 11-13; J-M 7; C-I 6, 12

Also used in: Sheet: J-I 5, 9, 15, 20; J-II 4, 26, 28; C-II 1, 3, 18-19

Question 16

What is the value of $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in case of a homogeneous equation in x and y?

Model Answer

$$\frac{dy}{dx} = \frac{y}{x}, \qquad \frac{d^2y}{dx^2} = 0$$

A homogeneous equation (all terms of the same degree, $= 0$) is a pair of lines $y = mx$ through the origin.

Ex: $ax^2 + 2hxy + by^2 = 0$: $y' = -\frac{ax+hy}{hx+by} = \frac{y}{x}$.

Grade - Low

Practice map, main concept: Sheet: J-I 25

Also used in: none

Question 17

How to differentiate an infinite expression such as $y = x^{x^{x^{\cdots}}}$ or $y = \sqrt{x + \sqrt{x + \cdots}}$?

Model Answer

The tail is the same as the whole: replace it by y, which gives a finite implicit relation; then differentiate (take ln for towers).

Ex: $y = x^y \Rightarrow \ln y = y \ln x \Rightarrow \frac{dy}{dx} = \frac{y^2}{x(1-y\ln x)}$; $y = \sqrt{x+y} \Rightarrow y' = \frac{1}{2y-1}$.

Grade - Low

Practice map, main concept: Sheet: J-I 18-20, 24

Also used in: none

Derivative of inverse of a function

Question 18

How to find the derivative of the inverse of a given function?

Model Answer

$$g = f^{-1}: \ g(f(x)) = x \Rightarrow g'(f(x)) = \frac{1}{f'(x)}; \qquad g'(b) = \frac{1}{f'(a)}, \ f(a) = b$$

To find $g'(b)$: solve $f(a) = b$ for a (usually by inspection), then $g'(b) = \frac{1}{f'(a)}$. Also $g''(b) = -\frac{f''(a)}{(f'(a))^3}$.

Ex: $f = x^5 + x^3$, $f(1) = 2$, $f'(1) = 8 \Rightarrow g'(2) = \frac{1}{8}$.

Grade - High

Practice map, main concept: Sheet: J-I 35-37; J-II 24-25; J-M 17-18; J-A 2; C-II 14 | Question Bank: 133

Also used in: none

Higher order derivatives

Question 19

What are the different ways to represent $\frac{dy}{dx}$, $\frac{d^2y}{dx^2}$, $\frac{d^3y}{dx^3}$, ...?

Model Answer

$$\frac{dy}{dx} = y' = y_1 = f'(x) = Dy; \quad \frac{d^2y}{dx^2} = y'' = y_2 = f''(x) = D^2y$$

$$\frac{d^3y}{dx^3} = y''' = y_3 = f'''(x) = D^3y; \quad \frac{d^ny}{dx^n} = y_n = f^{(n)}(x)$$

Grade - Low

Practice map, main concept: none

Also used in: none

Question 20

What do you mean by $\frac{d^2y}{dx^2}$?

Model Answer

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right), \qquad \frac{d^3y}{dx^3} = \frac{d}{dx}\left(\frac{d^2y}{dx^2}\right)$$

y' gives the slope, y'' the concavity of the graph. To prove a relation in y, y_1, y_2: differentiate the simplest (often squared or cleared) form, not y itself.

Ex: $y = \sin x \Rightarrow y_2 + y = 0$; $y = \sin^{-1} x \Rightarrow (1 - x^2)y_2 - xy_1 = 0$.

Grade - High

Practice map, main concept: Sheet: J-I 31, 38-44; J-II 26, 28, 43; J-M 2, 8, 11, 19, 21, 23-24; C-I 1, 8, 10; C-II 1-3, 5, 16-19 | Question Bank: 136-137, 139

Also used in: Sheet: J-I 45; J-II 29, 42; J-M 1, 5-6, 15; C-I 7; C-II 4, 6, 10, 13-14, 25 | Question Bank: 132, 134, 143, 145

Question 21

How to find $\frac{d^2y}{dx^2}$ when x and y are given parametrically?

Model Answer

$$\frac{d^2y}{dx^2} = \frac{d}{dt}\left(\frac{dy}{dx}\right)\cdot\frac{dt}{dx} = \frac{\dot{x}\,\ddot{y} - \dot{y}\,\ddot{x}}{\dot{x}^{\,3}}$$

Never divide $\frac{d^2y}{dt^2}$ by $\frac{d^2x}{dt^2}$.

Ex: $x = at^2,\ y = 2at$: $\frac{dy}{dx} = \frac{1}{t}$, $\frac{d^2y}{dx^2} = -\frac{1}{t^2}\cdot\frac{1}{2at} = -\frac{1}{2at^3}$.

Grade - Low

Practice map, main concept: Sheet: J-II 27; C-I 3, 17

Also used in: none

Question 22

How to find $\frac{d^2x}{dy^2}$ in terms of the derivatives of y w.r.t. x?

Model Answer

$$\frac{dx}{dy} = \frac{1}{y_1}, \qquad \frac{d^2x}{dy^2} = \frac{d}{dx}\left(\frac{1}{y_1}\right)\cdot\frac{dx}{dy} = -\frac{y_2}{y_1^3}$$

It is not $\frac{1}{y_2}$. The same result gives $g''(b) = -\frac{f''(a)}{(f'(a))^3}$ for $g = f^{-1}$.

Ex: $y = x^3 + x$: at $x = 1$, $y_1 = 4$, $y_2 = 6 \Rightarrow \frac{d^2x}{dy^2} = -\frac{6}{64} = -\frac{3}{32}$.

Grade - Low

Practice map, main concept: Sheet: J-I 46; C-II 6

Also used in: none

Analysis of ITF

Question 23

How to plot the graph of ITFs like $\sin^{-1}\frac{2x}{1+x^2}$ by defining them piecewise and then finding limits, intervals of monotonicity and nature of concavity?

Model Answer

Steps: domain and range; piecewise form (put $x = \tan\theta$ ...); limits at the ends and at the break points; f' for monotonicity (and points where f' does not exist); f'' for concavity; draw.

$$\sin^{-1}\frac{2x}{1+x^2} = \begin{cases} \pi - 2\tan^{-1}x, & x > 1 \\ 2\tan^{-1}x, & |x| \le 1 \\ -\pi - 2\tan^{-1}x, & x < -1 \end{cases}$$

Ex: so $f' = \frac{2}{1+x^2}$ for $|x| < 1$ and $-\frac{2}{1+x^2}$ for $|x| > 1$; $f'(1^-) = 1$, $f'(1^+) = -1$: corner at $x = \pm 1$.

Grade - Low

Practice map, main concept: Sheet: J-I 10, 16, 34; J-II 18-19, 46

Also used in: Sheet: J-I 17; J-II 14, 17; C-I 13; C-II 25

Miscellaneous

Question 24

What is the condition to differentiate a relation on both sides?

Model Answer

Only an identity (true for all x in an interval) can be differentiated and the result is again an identity. An equation true only at some points cannot.

Ex: $x^2 = 4$ holds only at $x = \pm 2$; differentiating gives $2x = 0$, which is false.

Grade - Low

Practice map, main concept: Sheet: J-II 22

Also used in: Sheet: J-I 31; C-II 4

Question 25

How to find the coefficient of x, x^2 in any expansion?

Model Answer

$$f(x) = a_0 + a_1 x + a_2 x^2 + \cdots \Rightarrow a_0 = f(0),\ a_1 = f'(0),\ a_2 = \frac{f''(0)}{2!},\ a_r = \frac{f^{(r)}(0)}{r!}$$

Ex: $f = (1+x)(1+x^2)(1+x^3)\ldots$: coefficient of x is $f'(0)$, found by log differentiation.

Differentiating an identity also gives sums: $(1+x)^n = \sum \binom{n}{r} x^r \Rightarrow \sum r\binom{n}{r} = n2^{n-1}$.

Grade - Low

Practice map, main concept: none

Also used in: none

Question 26

What is the condition for $f(x)$ to be divisible by $(x-a)^2$?

Model Answer

$$(x-a) \mid f \Rightarrow f(a) = 0; \quad (x-a)^2 \mid f \Rightarrow f(a) = f'(a) = 0$$
$$(x-a)^n \mid f \Rightarrow f(a) = f'(a) = \cdots = f^{(n-1)}(a) = 0$$

Ex: $x^3 + px + q$ divisible by $(x-1)^2$: $1 + p + q = 0$, $3 + p = 0 \Rightarrow p = -3,\ q = 2$.

Grade - Low

Practice map, main concept: Sheet: J-II 32

Also used in: Sheet: C-II 12

Question 27

How to differentiate a function containing modulus (or defined piecewise) at a point?

Model Answer

Near the given point find the sign of each expression inside $|\ |$, remove the modulus, then differentiate that piece. Where an inside expression is 0, check LHD and RHD.

$$\frac{d}{dx}|f(x)| = \frac{f(x)}{|f(x)|}\, f'(x), \quad f(x) \neq 0$$

Ex: $f = 2 + |x| - |x-1| - |x+1|$ near $x = \frac{1}{2}$: $f = 2 + x + (x-1) - (x+1) = x$, so $f'(\frac{1}{2}) = 1$.

Grade - Low

Practice map, main concept: Sheet: J-I 28, 30, 33

Also used in: none

Question 28

How to find $f(x)$ when its definition contains its own derivative values, e.g. $f(x) = x^3 + x^2 f'(1) + xf''(2) + f'''(3)$?

Model Answer

$f'(1)$, $f''(2)$, $f'''(3)$ are constants: call them a, b, c, differentiate repeatedly, put the points and solve the linear equations.

Ex: $f = x^3 + ax^2 + bx + c$: $c = f''' = 6$, $b = f''(2) = 12 + 2a$, $a = f'(1) = 3 + 2a + b \Rightarrow a = -5$, $b = 2$, $c = 6$.

Grade - High

Practice map, main concept: Sheet: J-II 29, 42; J-M 1, 6, 15 | Question Bank: 145

Also used in: none

Differentiation of Determinants

Question 29

How to differentiate a determinant without expansion?

Model Answer

Differentiate one row at a time, keep the other rows unchanged, and add the determinants (the same can be done column-wise).

$$\frac{d}{dx}\begin{vmatrix} f & g & h \\ u & v & w \\ l & m & n \end{vmatrix} = \begin{vmatrix} f' & g' & h' \\ u & v & w \\ l & m & n \end{vmatrix} + \begin{vmatrix} f & g & h \\ u' & v' & w' \\ l & m & n \end{vmatrix} + \begin{vmatrix} f & g & h \\ u & v & w \\ l' & m' & n' \end{vmatrix}$$

Ex: rows $(x^2, x^3, x^4), (2x, 3x^2, 4x^3), (1, 2, 3)$: the first term has two equal rows and the third row is constant, so $f' = \begin{vmatrix} x^2 & x^3 & x^4 \\ 2 & 6x & 12x^2 \\ 1 & 2 & 3 \end{vmatrix}$.

Grade - High

Practice map, main concept: Sheet: J-II 30-31; J-M 12-13; C-II 10 | Question Bank: 132, 140-141

Also used in: Sheet: J-II 32

L'Hospital Rule

Question 30

What is L'Hospital rule?

Model Answer

$$\lim_{x \to a} \frac{f(x)}{g(x)} = \lim_{x \to a} \frac{f'(x)}{g'(x)} \quad \left(\frac{0}{0} \text{ or } \frac{\infty}{\infty}\right)$$

Ex: $\lim_{x \to 0} \frac{1 - \cos x}{x^2} = \lim_{x \to 0} \frac{\sin x}{2x} = \frac{1}{2}$.

Grade - Medium

Practice map, main concept: Sheet: J-I 47-48, 50; J-II 33-35, 37-41; J-M 25; J-A 6 | Question Bank: 144

Also used in: Sheet: J-I 51; J-II 48; C-II 11-12

Question 31
What are the things to keep in mind while applying L'Hospital rule?

Model Answer
(i) Only for $\frac{0}{0}$ or $\frac{\infty}{\infty}$ (convert $0\cdot\infty$, $\infty-\infty$ first); (ii) f, g differentiable near a, $g' \neq 0$; (iii) differentiate numerator and denominator separately, NOT the quotient rule; (iv) a may be finite or $\pm\infty$.

Simplify with standard limits ($\frac{\sin x}{x} \to 1$) between steps to keep the work short.

Grade - **Low**
Practice map, main concept: none
Also used in: none

Question 32
How many times can we apply L'Hospital rule?

Model Answer
Any number of times, as long as the new limit is still $\frac{0}{0}$ or $\frac{\infty}{\infty}$; check the form before every step.

Ex: $\lim_{x\to\infty} \frac{x^{100}}{e^x}$: after 100 steps $\lim \frac{100!}{e^x} = 0$.

Grade - **Medium**
Practice map, main concept: Sheet: C-II 12
Also used in: Sheet: J-I 47, 49; J-II 34, 36; J-A 6

Question 33
How to find the limit in case of 0^0, ∞^0 forms?

Model Answer

$$y = f^{\,g} \Rightarrow \ln y = g \ln f = \frac{\ln f}{1/g} \ \left(\frac{\infty}{\infty}\right); \qquad \lim f^{\,g} = e^{\lim g \ln f}$$

Ex: $\lim_{x\to 0^+} x^x$: $\lim x \ln x = \lim \frac{\ln x}{1/x} = \lim(-x) = 0 \Rightarrow$ limit $= e^0 = 1$.

Grade - **Low**
Practice map, main concept: Sheet: J-I 49, 51; J-II 36, 48; C-II 11
Also used in: none

Differentiation of Integration

Question 34
How to differentiate an indefinite integration without integrating it first?

Model Answer

$$I = \int f(x)\,dx \Rightarrow \frac{dI}{dx} = f(x)$$

Ex: $\frac{d}{dx}\int e^{x^2}\,dx = e^{x^2}$ (no antiderivative needed).

Grade - **Low**
Practice map, main concept: Sheet: C-I 9
Also used in: none

Question 35

How to differentiate a definite integration without integrating it first with the help of Leibnitz rule?

Model Answer

$$\frac{d}{dx}\int_{g(x)}^{h(x)} f(t)\,dt = f(h(x))\,h'(x) - f(g(x))\,g'(x) \quad (f \text{ continuous})$$

If x also occurs inside the integrand, take it out first (e.g. $\int_0^x (x-t)f(t)dt = x\int_0^x f - \int_0^x tf$).

Ex: $\frac{d}{dx}\int_x^{x^2} \frac{dt}{\ln t} = \frac{2x}{2\ln x} - \frac{1}{\ln x} = \frac{x-1}{\ln x}$.

Grade - **Low**

Practice map, main concept: none

Also used in: none